

MATHEMATICS(SM015)

SEMESTER 1

SPECIMEN SM015_____

2 HOURS

1. This question booklet consists of 10 questions.
 2. Answer **ALL QUESTIONS**
 3. All steps must be shown clearly.
 4. Numerical answers can be given in the form of π, e , surd, fractions or up to three significant figures.
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Differentiation and Integration :

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

Sphere : $V = \frac{4}{3}\pi r^3$

$$S = 4\pi r^2$$

Right circular cone : $V = \frac{1}{3}\pi r^2 h$

$$S = \pi r s$$

Right circular cylinder : $V = \pi r^2 h$

$$S = 2\pi r h$$

LIST OF MATHEMATICAL FORMULAE

Trigonometry : $\sin (A \pm B) = \sin A \cos B \pm \cos A \sin B$

$$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2\cos^2 A - 1$$

$$= 1 - 2\sin^2 A$$

$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

Answer all questions

SECTION A

This section consists of 3 questions. Answer all questions.

- 1 (a) Evaluate $\lim_{x \rightarrow \infty} \frac{2x^2 + 8x - 2}{x^2 + 1}$. [3 marks]
- b) Determine the values of H and K for such that the following function is continuous at $x = -1$ and $x = 1$.
- $$f(x) = \begin{cases} Hx + K, & x \leq -1 \\ 4x^2 - 5Hx - K, & -1 < x \leq 1 \\ -4, & x > 1 \end{cases}$$
- [6 marks]
- 2 (a) Find the derivative of $f(x) = \frac{2}{3\sqrt{x}}$ using the first principles. [5 marks]
- (b) It is given that $x^2y - \sin(2x^2 + e^2) = 0$. Show that $2y(1 + 8x^4) + 4xy' + x^2y'' - 4\cos(2x^2 + e^2) = 0$. [6 marks]
- 3 Water is poured into vase at a constant rate of $80\pi \text{ cm}^3\text{s}^{-1}$. When the depth of water is h cm, the volume of water, $V \text{ cm}^3$ is given by $V = 4\pi h(h + 4)$. Find the rate of change of the depth of water in the vase when $h = 6$ cm. [5 marks]

SECTION B

- 1 The complex numbers Z and W are given by $Z = 2 - i$ and $W = \frac{i}{1-i}$

Find $Z+W$ in the form of $a+bi$. Hence, express $Z+W$ in polar form.

[8 marks]

- 2 (a) Solve the inequality $\frac{x+1}{x-2} \geq 3$

[4 marks]

(b) Find the set of values of x for which $2 + \left| \frac{x}{4+x} \right| < 4$

[6 marks]

- 3 Given $p(x) = e^{-2x}$, $x \in \mathbb{R}$ and $q(x) = 1 + x^3$, $x \in \mathbb{R}$

(a) Find $p^{-1}(x)$

(b) Find $(q \circ p)(x)$.

(c) Solve $(p \circ q)(x) = e$.

[7 marks]

- 4 (a) Function f is defined as follows :

$$f(x) = \begin{cases} e^x & , x \geq 0 \\ 1 - 9x^2 & , x < 0 \end{cases}$$

(i) Find $\lim_{x \rightarrow 0} f(x)$

(ii) Determine whether f is continuous at $x = 0$. Give reason for your answer.

[6 marks]

- (b) Determine the value of p so that the function

$$f(x) = \begin{cases} 2x^2 + 8 & , x \leq 2 \\ 4px & , x > 2 \end{cases} \text{ is continuous at } x = 2$$

[5 marks]

5. (a) Find $\frac{dy}{dx}$ if $y = \ln \sqrt{\frac{3+x}{3-x}}$. [5 marks]

(b) Given $x^2 + 4y^2 - 5xy = 0$. Find y' when $x = 1$ [7 marks]

6 Given that $x = \frac{1}{t+1}$ and $y = \frac{t^2}{t+1}$.

(a) Find $\frac{dy}{dx}$ in terms of t .

(b) Find $\frac{d^2y}{dx^2}$ at the point $\left(\frac{1}{2}, \frac{1}{2}\right)$

[12 marks]

7. (a) Find the local minimum and the local maximum of the curve $y = x + \frac{9}{x}$.

[6 marks]

(b) A closed rectangular oil tank has a square base of length x m.

The total surface area of the oil tank is $96m^2$

(i) Show that the height of the oil tank is $\left(\frac{24}{x} - \frac{1}{2}x\right)m$.

(ii) Find the maximum volume of the oil tank.

[9 marks]

END OF QUESTIONS